

Shi Boao

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EDUCATION

The University of Hong Kong <i>Bachelor of Engineering, major in Computer Science, minor in Math&Stat</i>	Sept. 2022 - Present GPA: 4.12/4.3 Rank: 2/140+
University of Illinois at Urbana-Champaign (Semester exchange) <i>The Grainger College of Engineering</i>	2025 Spring
Hangzhou No.2 High School of Zhejiang Province	Sept. 2019 - June. 2022

RELEVANT COURSEWORK

Grades: 14 A+ out of 17 major courses
Courses: Computer Graphics: A+; Linear algebra A+; Probability & Statistics I: A+; Calculus and ordinary differential equations: A+; Operating System: A+; Object-oriented Programming: A+; Discrete Math: A+; Data Structures and Algorithms: A+; Quantum information and computation: A+; Principles of programming languages: A+; etc.

SKILLS

Communication Languages: Mandarin: Native; English: Fluent (TOEFL iBT: **104** (with sub-scores 24+))
Main Programming Languages: Python $\approx$ C/C++ > Java
Full Stack Developing: HTML, CSS, JavaScripts(JQuery, ReactJS, NodeJS), MySQL/MongoDB, PHP
Packages: Pytorch, ROS1, ROS2, IsaacGym, OpenCV, colmap, 3DGS
Software: Blender, MeshLab, Unity

PUBLICATIONS

EvoR: Evolving Retrieval for Code Generation
*Hongjin Su, Shuyang Jiang, Yuhang Lai, Haoyuan Wu, **Boao Shi**, Che Liu, Qian Liu, Tao Yu*
Accepted as The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP2024), Findings

INTERNSHIP

Research Intern in Infobodied Lab <i>Robotics, Computer Vision, Computer Graphics</i>	May. 2024 – Now
- Research in Prof. Yanchao Yang’s lab in HKU Musketeers Foundation Institute of Data Science, focusing on Embodied AI. I am leading a research project about humanoid scene interaction generation, which is prepared to submit ICCV2025 - Understanding the state of the art in embodied motion generation of humanoid robots, including motion capture, motion synthesis, and motion control - Build up an automatic human whole body motion data collection system, which extract demonstration data from real human video and reconstruct real scenes into their digital twins; reconstruct interacted objects and track their 6D pose; collecting the largest dataset to date for human-scene/object interaction - Using deep learning to train a text-driven scene-conditioned human motion generation model, which can be used as high-level motion guide for low-level humanoid robots control	
Research Assistant in XLang Lab <i>Natural language processing, LLM</i>	Dec. 2023 – Mar. 2024
- Participated in a research project about Information Retrieval and Code Generation - Built a benchmark for testing information retrieval model reasoning ability, including data collection, data preprocessing - Curated a dataset about long-tail programming Languages, testing LLM generalization ability when aided with retrieval argumentation generation - Outputted a research paper "EvoR: Evolving Retrieval for Code Generation"	

PROJECTS

Robot auto-aiming system <i>Robotics, Computer Vision, Control Theory, Sensors</i>	May. 2023 – Apr. 2024
- Using traditional algorithm(OpenCV) or Yolo model to recognize lights on armors - Trained a shallow layer neural network for fast recognizing number on armors - Use PnP algorithm to estimate pose of target armor in 3D space and control robot to shoot through ROS - Using Extended Kalman Filter algorithm to predict, denoise, and track target's moving and spinning status	

EXPERIENCE & TEACHING

HKU Innovation Wing RoboMaster Team <i>Member</i>	Sept. 2022 - Apr. 2023
Leader of computer vision group, develop first generation of robot auto-aiming system Won Eighth-ranked team in both 3V3 and Infantry Match in RoboMaster University League (RMUL) held by DJI	
HKU ENGG1330 (programming I) <i>Teaching Assistant</i>	Sept. 2023 – Dec. 2023
As student teaching assistant for 2-hours tutorial every week	
HKU COMP2396 (Object-oriented Programming and Java) <i>Teaching Assistant</i>	Sept. 2024 – Dec. 2024
As student teaching assistant for tutorial questions answering	

COMPETITON AWARDS

Huawei Coding Challenge	March 2023
Rank: 25/311, Third Prize	
2022 International Collegiate Programming Contest Asia East Continent Final Contest	March 2023
Bronze Medal	
2022 International Collegiate Programming Contest Hong Kong Regional Contest	Jan. 2023
Rank: 19, Silver Medal	
National Olympiad in Informatics in Provinces(NOIP) (Zhejiang Province)	2018 - 2020
First Prize × 3 times	

TALKS

HKU CSSAUD Senior Academic Sharing Session (Engineering)	Sept. 2024
Sharing study experience in computer science, and how to begin research in AI. Record & PPT	
RoboMaster Team Training Workshop - Introduction to Linux System (Algorithm group)	Sept. 2023
Introduction to Linux System, system overview, basic command line usage, remote server usage	

SCHOLARSHIP & AWARDS

Hong Kong University Alumni Prize	Oct. 2024
Academic-merit-based Award, The University of Hong Kong (1 awardee in Faculty of Engineering)	
Talent Development Scholarship	June. 2024
HK\$10,000, Recognize talent in scope of Innovation, science and technology, The Hong Kong Government	
Dean's Honours List	Mar. 2024
Academic-merit-based Award for Engineering students in HKU (Top 10%), The University of Hong Kong	
Young Tsun Dart Scholarship	Nov.2023
HK\$15,000, Academic-merit-based Scholarship for Engineering students in HKU (Top 2 in Faculty of Engineering)	
China Soong Ching-ling Foundation Zhiyuan Scholarship	2022-2026
¥80,000 annually, Merit-based Scholarship for students pursuing Bachelor Degree in HK (~ 30 awardees nation-wide)	